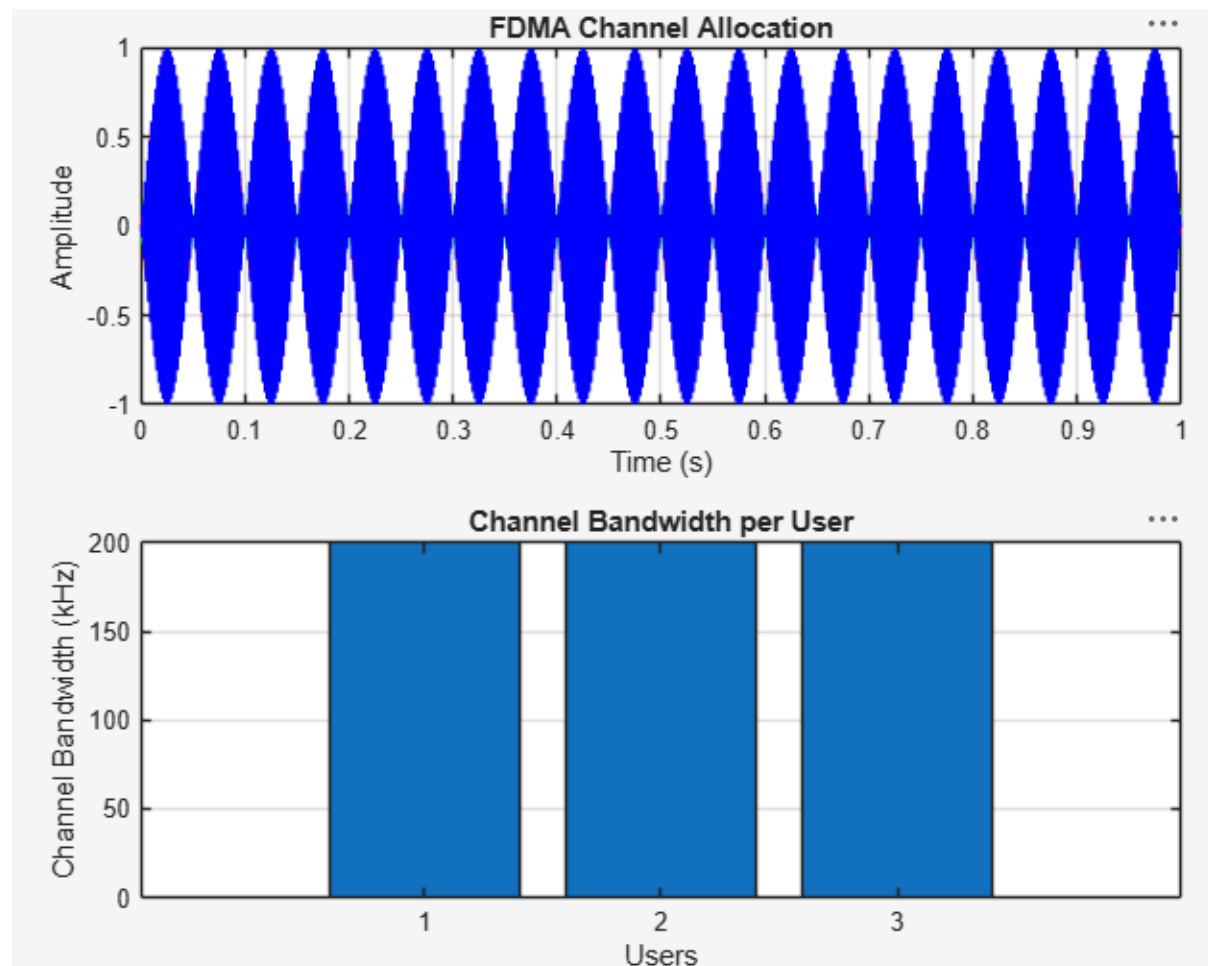


## Ex.No:1: Simulation of FDMA in MATLAB

### Objective 1: Matlab Code and Output

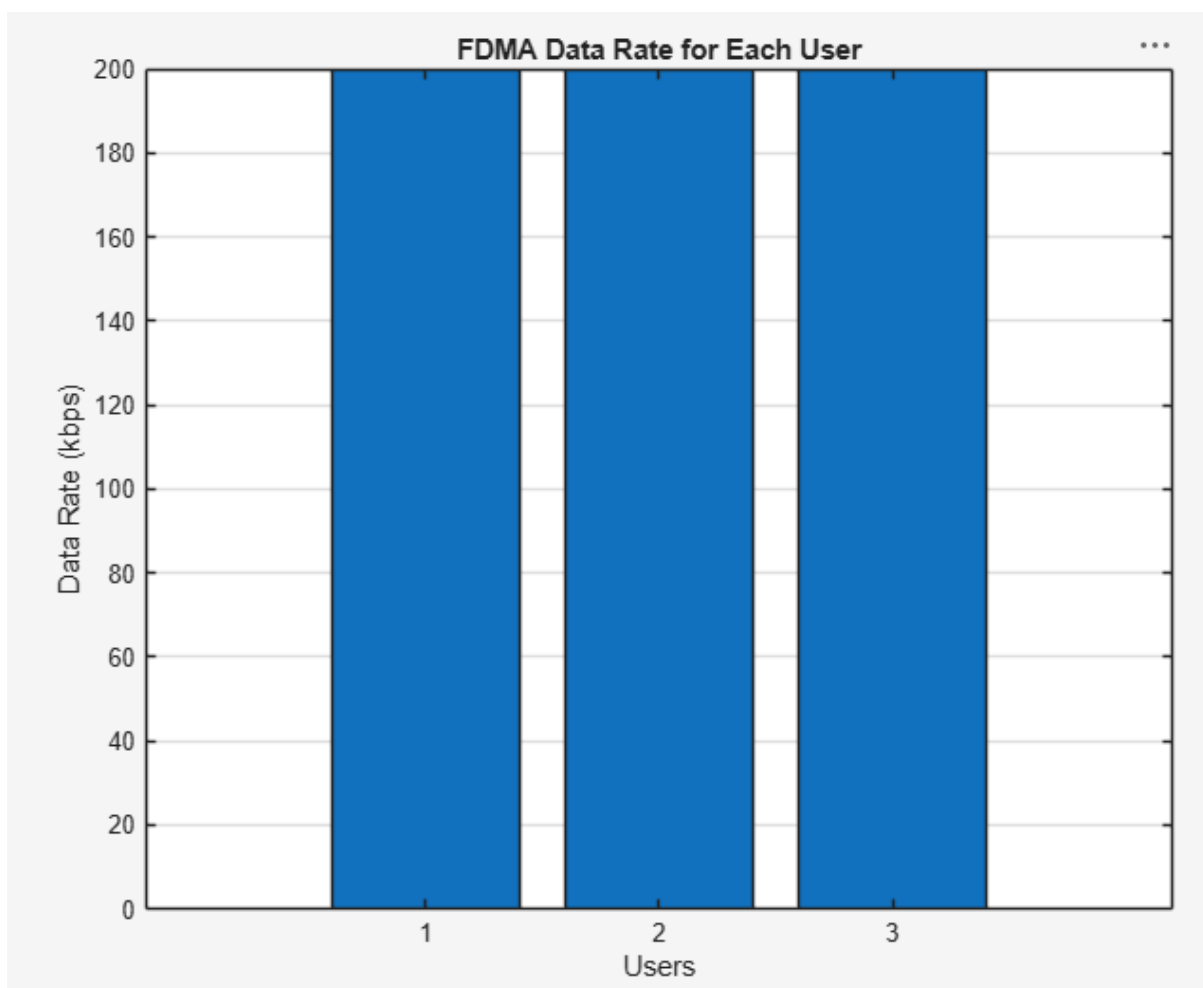
/MATLAB Drive/slotb1.m

```
1  clc; clear; close all;
2  fs=1000; t=0:1/fs:1;
3  TBW=600; Users=3;
4
5  CBW=TBW/Users;
6  fc=[100 300 500];
7  m=sin(2*pi*10*t);
8  s1=m.*cos(2*pi*fc(1)*t);
9  s2=m.*cos(2*pi*fc(2)*t);
10 s3=m.*cos(2*pi*fc(3)*t);
11 subplot(2,1,1)
12 plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
13 title('FDMA Channel Allocation'),xlabel('Time (s)'),ylabel('Amplitude')
14 subplot(2,1,2)
15 bar(1:Users,CBW*ones(1,Users)),grid on
16 xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per
```



## Objective 2: Matlab Code and Output

```
18 BW = 200; % Channel Bandwidth (kHz)
19 Users = 3;
20 M = 2; % BPSK Modulation
21 DataRate = BW*log2(M); % Data Rate (kbps)
22 Rate = DataRate*ones(1,Users);
23 disp(' User Data Rate (kbps)')
24 disp([(1:Users)' Rate'])
25 figure
26 bar(1:Users,Rate)
27 xlabel('Users')
28 ylabel('Data Rate (kbps)')
29 title('FDMA Data Rate for Each User')
30 grid on
```



### Objective 3: Matlab Code and Output

```
32 Ps = 1; % Signal Power (W)
33 Pn = [0.1 0.05 0.01]; % Noise Power (W)
34 Users = 1:3;
35 SNR = 10*log10(Ps./Pn); % SNR in dB
36 disp('User Noise Power(W) SNR(dB)')
37 disp([Users' Pn' SNR'])
38 figure
39 bar(Users,SNR)
40 xlabel('Users')
41 ylabel('SNR (dB)')
42 title('Signal-to-Noise Ratio of FDMA Users')
43 grid on
44
45
```

